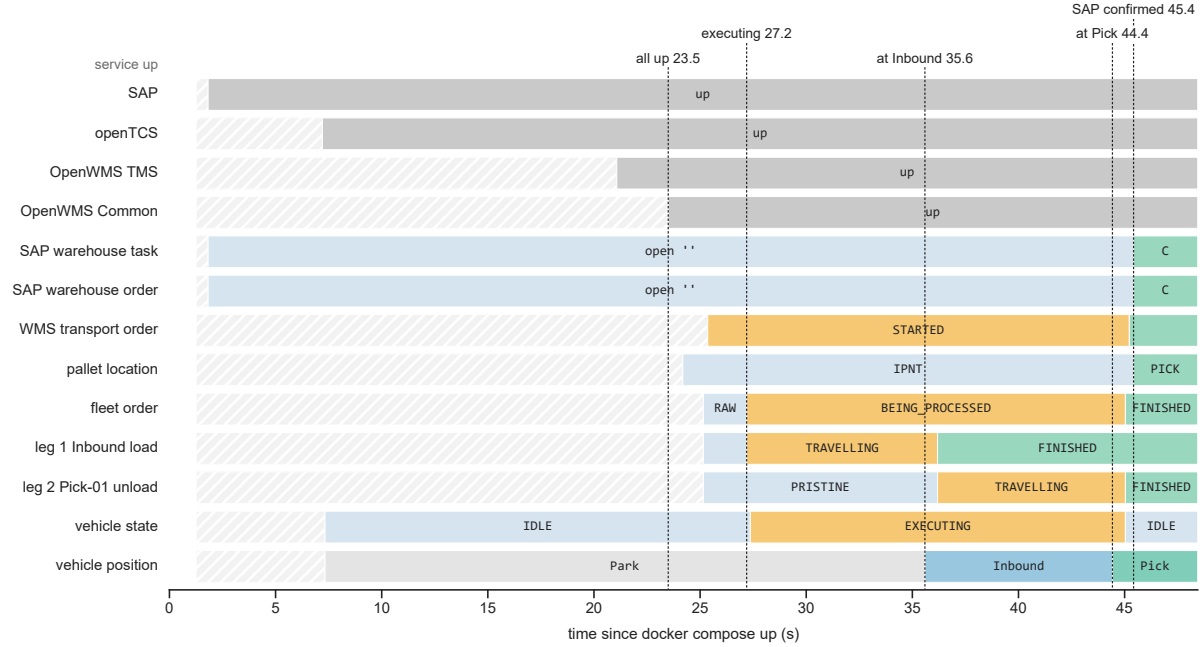
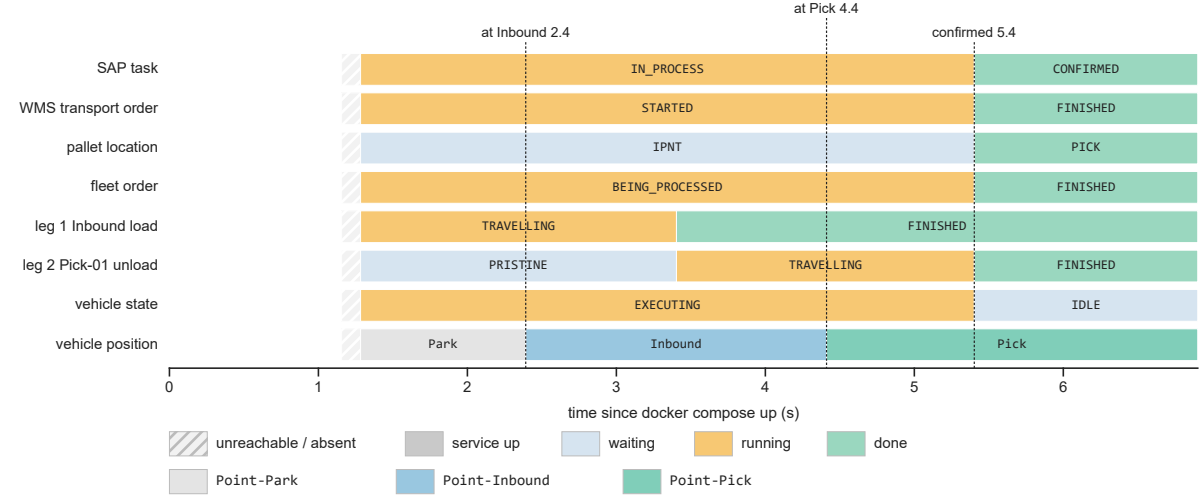
a **Full stack: SAP CAP (EWM API) + OpenWMS + openTCS**



b **Python simplified chain (same docker compose up)**



Data: docs/figures/run_trace.json, run started 2026-11-08 20:39:23, 231 samples, median interval 0.2 s, max 0.8 s.

Bridge journal (full stack): task taken to SAP confirmation 21.0 s. The Python bridge took the task before the first sample, hence IN_PROCESS from the start.